

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XIII

JEE (Main)-2022

TEST DATE: 31-05-2022

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 Questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Physics

PART – A

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. The velocity of a particle depends upon the time t according to the equation:

$$v = \sqrt{ab} + bt + \frac{c}{d+t}$$

The physical quantities which are represented by a , b , c and d , are in the following order:

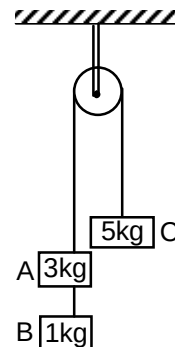
- (A) distance, distance, acceleration, time
(B) distance, acceleration, distance, time
(C) acceleration, distance, distance, time
(D) none of the above

2. Dimensions of electrical resistance is:

- (A) $[ML^2T^{-3}A^{-1}]$
(B) $[ML^2T^{-3}A^{-2}]$
(C) $[ML^3T^{-3}A^{-2}]$
(D) $[ML^{-1}T^3A^2]$

3. In the adjoining figure, the tension in the string connecting A and B is

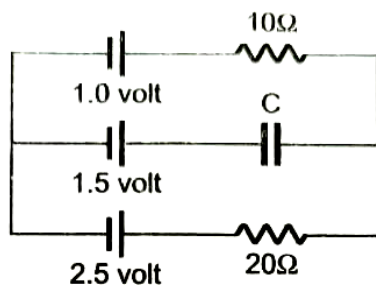
- (A) g
(B) $\frac{g}{9}$
(C) $\frac{8g}{9}$
(D) $\frac{10g}{9}$



4. An object of mass 5 kg falls from rest through a vertical distance of 20 m and reaches ground with a velocity of 10 m/s. The work done by air resistance ($g = 9.8 \text{ ms}^{-2}$) is:
- (A) +730 J
(B) -730 J
(C) +980 J
(D) -980 J

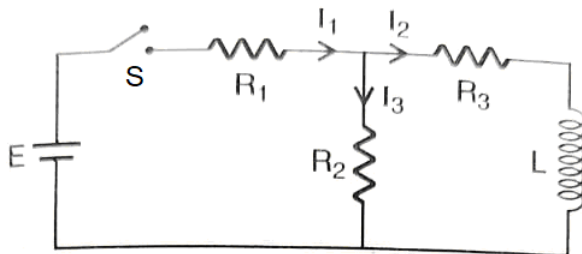
5. An explosion blows a rock into three parts. Two parts go off at right angles to each other. These two are 1 kg first part moving with a velocity of 12 m s^{-1} and 2 kg second part moving with a velocity 8 ms^{-1} . If the third part flies-off with a velocity of 4 ms^{-1} , its mass would be:
- (A) 7 kg
(B) 17 kg
(C) 3 kg
(D) 5 kg
6. A satellite orbiting around the earth of a radius R is shifted to an orbit of radius $2R$. How many times the time taken for one revolution increase?
- (A) 8 times
(B) 2 times
(C) 2.5 times
(D) 2.8 times
7. When water flows at a rate Q through a tube of radius r placed horizontally, a pressure difference p develops across the ends of the tube. If the radius of the tube is doubled and the rate of flow halved, the pressure difference will be:
- (A) $8p$
(B) p
(C) $p/8$
(D) $p/32$
8. Oxygen is 16 times heavier than hydrogen. Equal volumes of hydrogen and oxygen are mixed. The ratio of the velocity of sound in the mixture to that in oxygen is:
- (A) $\sqrt{\frac{1}{8}}$
(B) $\sqrt{\frac{32}{17}}$
(C) $\sqrt{\frac{17}{32}}$
(D) $\sqrt{8}$
9. A sound absorber attenuates the sound level by 20 dB. The intensity decreases by a factor of:
- (A) 100
(B) 1000
(C) 10000
(D) 10
10. The critical angle for light going from medium x into medium y is θ . The speed of light in medium x is v . The speed of light in medium y is:
- (A) $v(1 - \cos\theta)$
(B) $v / \sin \theta$
(C) $v / \cos \theta$
(D) $v \cos \theta$

11. A point source of light is placed 4 m below the surface of water of refractive index $5/3$. The minimum diameter of a disc, which should be placed over the source, on the surface of water to cut-off all light coming out of water is:
- (A) 1 m
(B) 4 m
(C) 3 m
(D) 6 m
12. Concave and convex lenses are placed touching each other. The ratio of magnitudes of their power is 2 : 3. The focal length of the system is 30 cm. Then the focal lengths of individual lenses are:
- (A) -75, 50
(B) -15, 10
(C) 75, 50
(D) 75, -50
13. A prism of a certain angle deviates the red and blue rays by 8° and 12° respectively. Another prism of the same angle deviates the red and blue rays by 10° and 14° respectively. The prisms are small angled and made of different materials. The dispersive powers of the materials of the prisms are in the ratio:
- (A) 5 : 6
(B) 9 : 11
(C) 6 : 5
(D) 11 : 9
14. The current gain of a transistor in common base mode is 0.995. The current gain of the same transistor in common emitter mode is:
- (A) 197
(B) 201
(C) 198
(D) 199
15. In the following circuit diagram the potential difference across the plates of the capacitor C is:

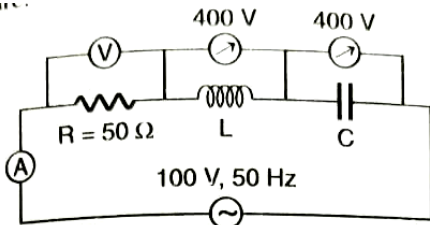


- (A) 2.5 volt
(B) 1.5 volt
(C) 1.0 volt
(D) zero

16. The resistivity of a potentiometer wire is $40 \times 10^{-8} \text{ ohm-m}$ and its area of cross-section is $8 \times 10^{-6} \text{ m}^2$. If 0.2 ampere current is flowing through the wire, the potential gradient will be:
- (A) 10^{-2} V/m
 (B) 10^{-1} V/m
 (C) $3.2 \times 10^{-2} \text{ V/m}$
 (D) 1 V/m
17. In the circuit shown in the following figure $E = 10\text{V}$, $R_1 = 2 \text{ ohm}$, $R_2 = 3 \text{ ohm}$ and $R_3 = 6 \text{ ohm}$ and $L = 5 \text{ henry}$. The current I_1 just after pressing the switch S is:



- (A) $(10/4) \text{ amp}$
 (B) $(10/5) \text{ amp}$
 (C) $(10/12) \text{ amp}$
 (D) $(10/6) \text{ amp}$
18. In the series LCR circuit, the voltmeter and ammeter readings are:

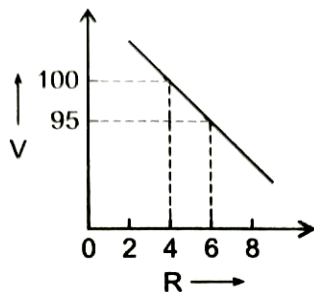


- (A) $V = 100 \text{ volt}$, $I = 2 \text{ amp}$
 (B) $V = 100 \text{ volt}$, $I = 5 \text{ amp}$
 (C) $V = 1000 \text{ volt}$, $I = 2 \text{ amp}$
 (D) $V = 300 \text{ volt}$, $I = 1 \text{ amp}$
19. The ratio of the frequencies of the long wavelength limits of the Lyman and Balmer series of hydrogen is:
- (A) $27 : 5$
 (B) $5 : 27$
 (C) $4 : 1$
 (D) $1 : 4$
20. The wavelength of K_{α} X-rays produced by an X-ray tube is $1.785 \text{ \AA}$. Find the atomic number of the anode material of the tube ($R = 109737 \text{ cm}^{-1}$):
- (A) 24
 (B) 32
 (C) 48
 (D) 27

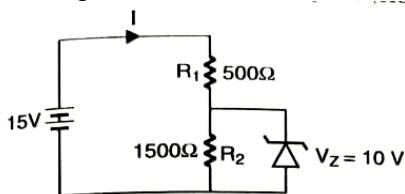
SECTION – B (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXX.XX).

21. The Poisson's ratio of a material is 0.4. If a force is applied to a wire of this material, there is a decrease of cross-sectional area by 2%. The percentage increase in its length is (in percentage)
22. A body of mass 2 kg is floating in water with half its volume submerged. What would be the force required to wholly submerge it into water? (in N) (take $g = 9.8 \text{ m/s}^2$)
23. If the temperature of a hot body is raised by 5%, then the heat energy radiated would increase by: (in percentage)
24. A sample of ideal gas ($\gamma = 1.4$) is heated at constant pressure. If an amount of 100 J is supplied to the gas, the work done by the gas is: (in Joule)
25. The variation of potential with distance R from a fixed point is shown in figure. The electric field at $R = 5 \text{ m}$ is:



26. The width of a single slit, if the first minimum is observed at an angle of 2° with a light of wavelength $6980 \text{ \AA}$, (in mm) is:
27. The balancing length for a cell is 560 cm in a potentiometer experiment. When an external resistance of 10Ω is connected in parallel to the cell, the balancing length changes by 60 cm. The internal resistance of the cell (in ohm) is:
28. A plane electromagnetic wave, $E_z = 100 \cos(6 \times 10^8 t + 4x) \text{ V/m}$ propagates in a medium. Find out refractive index of medium.
29. In the given circuit the current through the Zener diode is: (in mA)



30. If per nucleon binding energies for ${}_3\text{Li}^7$ and ${}_2\text{He}^4$ are 5.60 MeV and 7.06 MeV, then the energy released in the nuclear process: (in MeV)
 ${}_3\text{Li}^7 + {}_1\text{p}^1 \rightarrow 2 {}_2\text{He}^4$ is:

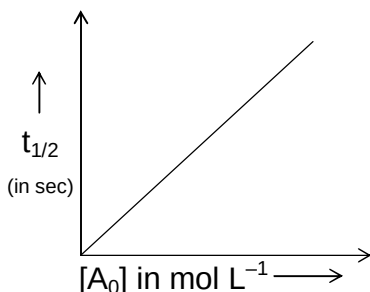
Chemistry

PART – B

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31.

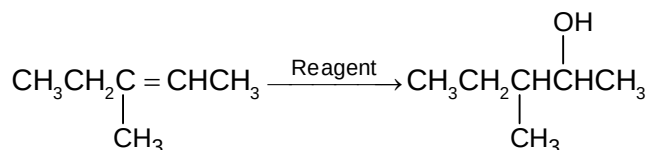


What is the unit of rate constant of the reaction, for which the above graph is given?

- (A) s^{-1}
 - (B) $\text{mol L}^{-1}\text{s}^{-1}$
 - (C) $\text{mol L}^{-1}\text{s}$
 - (D) $\text{mol}^{-1} \text{L}^2\text{s}^{-1}$
32. The soda extract of which of the following compound gives the test of nitrogen?
- (A) NH_2NH_2
 - (B) HNO_2
 - (C) HCONH_2
 - (D) All are correct
33. Which of the following reaction produces Cl_2 gas?
- (A) $\text{NaCl} + \text{H}_2\text{SO}_4 \longrightarrow$
 - (B) $\text{KClO}_3 \xrightarrow{\text{Heat}}$
 - (C) $\text{HCl} + \text{O}_3 \longrightarrow$
 - (D) $\text{KCl} + \text{AgNO}_3 \longrightarrow$
34. Which of the following metallurgical process is not needed in order to extract Zn from ZnCO_3 ?
- (A) Calcination
 - (B) Roasting
 - (C) Carbon reduction
 - (D) Distillation
35. Which of the following salt can increase the boiling point of water by maximum extent? (Assume equimolar quantities are added)
(Assume complete dissociation of the salts)
- (A) NaCl
 - (B) CaCl_2
 - (C) AlCl_3
 - (D) KCl

36. The heat of atomization of a compound XY_3 in gaseous state is $E \text{ kJ mol}^{-1}$. What is the $X - Y$ bond energy in kJ mol^{-1} unit?
- (A) E
 (B) $3E$
 (C) $\frac{E}{3}$
 (D) $\frac{E}{6}$
37. In aqueous solution beryllium sulphate exists as:
- (A) $[\text{Be}(\text{H}_2\text{O})_4]\text{SO}_4$
 (B) $[\text{Be}(\text{H}_2\text{O})_6]\text{SO}_4$
 (C) $[\text{Be}(\text{H}_2\text{O})_3]\text{SO}_4$
 (D) $[\text{Be}(\text{H}_2\text{O})_5]\text{SO}_4$
38. The normal boiling point of a liquid 'X' is 500 K. Which of the following statement(s) is incorrect for the following process?
 $X(\text{liquid}) \longrightarrow X(\text{gas})$
- (A) At 500 K and 1 atm pressure, $\Delta G = 0$
 (B) At 500 K and 1.5 atm pressure, $\Delta G = +ve$
 (C) At 500 K and 0.2 atm pressure, $\Delta G = -ve$
 (D) At 520 K and 1 atm pressure, $\Delta G = +ve$
39. In the crystal lattice of CsCl , Cs^+ ions are present at the
- (A) tetrahedral voids of the unit cell
 (B) octahedral voids of the unit cell
 (C) body-centre of the unit cell
 (D) face-centre of the unit cell
40. A weak acid HX has the dissociation constant $1 \times 10^{-5} \text{ M}$. It forms a salt NaX on reaction with alkali. The percentage hydrolysis of 0.1 M solution of NaX is
- (A) 0.0001%
 (B) 0.01%
 (C) 0.1%
 (D) 0.15%
41. What is the molar solubility of $\text{Mn}(\text{OH})_2$ ($K_{sp} = 4.5 \times 10^{-14}$) in a buffer solution containing equal amounts of NH_4^+ & NH_3 ($K_b = 1.8 \times 10^{-5}$)
- (A) 3.0×10^{-5}
 (B) 1.38×10^{-4}
 (C) 1.38×10^{-2}
 (D) 7.3×10^{-6}
42. Which of the following reagent can convert phenol (PhOH) to anisole (PhOCH_3)?
- (A) CHCl_3
 (B) $\text{CH}_2\text{N}_2/\Delta$
 (C) CH_3Cl
 (D) HCHO

43.



Which reagent forms the above organic product as the only product in the above reaction?

- (A) $\text{H}_2\text{O}/\text{H}^+$
 (B) $\frac{\text{Hg}(\text{OCOCH}_3)_2, \text{H}_2\text{O}}{\text{NaBH}_4}$
 (C) $\frac{\text{B}_2\text{H}_6, \text{THF}}{\text{NaOH}, \text{H}_2\text{O}_2}$
 (D) $\text{KMnO}_4/\text{OH}^-/\text{Cold}$

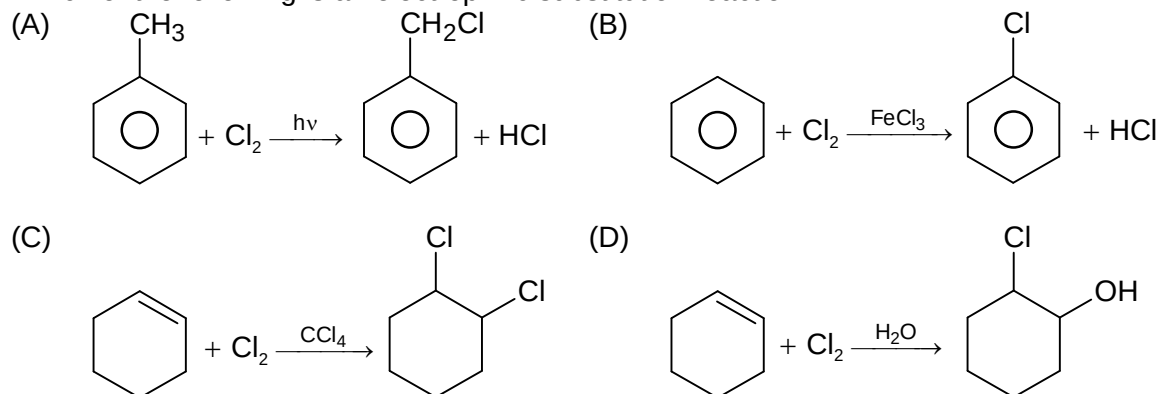
44. Which of the following compound forms two stereoisomers when treated with NH_2OH in weakly acidic medium?

- (A) CH_3COCH_3
 (B) HCHO
 (C) $\text{CH}_3\text{COC}_2\text{H}_5$
 (D) $\text{C}_2\text{H}_5\text{COCH}_2\text{CH}_3$

45. Which complex cannot form precipitate with AgNO_3 solution?

- (A) $\text{Na}_2[\text{ZnCl}_4]$
 (B) $[\text{Cr}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
 (C) $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$
 (D) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$

46. Which of the following is an electrophilic substitution reaction?



47. In borax bead test, which compound is formed?

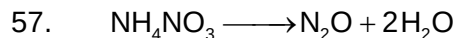
- (A) Meta borate
 (B) Ortho borate
 (C) Tetra borate
 (D) Double oxide

48. AlF_3 is most soluble in
 (A) HF
 (B) KF
 (C) KHF_2
 (D) F_2
49. $\text{SO}_3(\text{g}) + \text{CO}(\text{g}) \rightleftharpoons \text{SO}_2(\text{g}) + \text{CO}_2(\text{g})$
 Which of the following activity can reduce the equilibrium concentration of CO to half?
 (A) Increasing pressure twice
 (B) Increasing the volume of reaction container to two times of its initial value
 (C) Adding two moles of CO_2 gas
 (D) Adding two moles of helium gas at constant pressure
50. Which of the following is a paramagnetic metal ion?
 (A) Zn^{2+}
 (B) Ti^{4+}
 (C) Mn^{5+}
 (D) Cu^+

SECTION – B
(Numerical Answer Type)

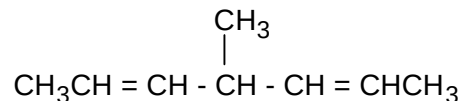
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51. An alkyl polychloride(X) on hydrolysis forms HCOOH . How many total number of atom(s) is/are present in the molecule(X)?
52. The pH of 0.1 M aqueous solution of the sodium salt of a weak monobasic acid(HA) is 12.8. What is the $\text{p}K_a$ value of the acid?
53. Two moles of glycol is dissolved in 891.24 g of water. The solution is cooled to -4.8°C . How many gram of ice is formed in the cooling process? [K_f of water = $1.86 \text{ K kg mol}^{-1}$]
54. $\text{X}(\text{g}) \rightleftharpoons \text{Y}(\text{g}) + \text{Z}(\text{g})$
 250 mole of 'X' is taken in a 100 L container at a certain temperature. 50 mole percentage of 'X' undergoes reaction at equilibrium. How many moles of 'X' should be added at equilibrium so that concentration of Y will become 1.3 M at the new equilibrium?
55. How much heat is absorbed in Kcal mol^{-1} unit by heating one mole of an ideal gas from 300 K to 1300 K at constant volume? $\left[C_v = \frac{3}{2} R \right]$ [$R = 2 \text{ cal K}^{-1} \text{ mol}^{-1}$]
56. $\text{Zn}(\text{s}) | \text{Zn}^{2+}(0.01 \text{ M}) || \text{H}^+ | \text{H}_2(1 \text{ atm}), \text{Pt}$
 The e.m.f of the above electrochemical cell is 0.6418 V. What is the pH of the cathode half-cell if $E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76 \text{ V}$



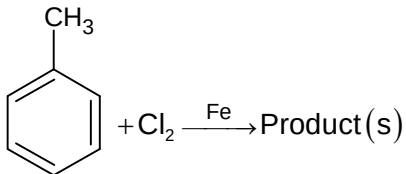
Let the molecular mass of NH_4NO_3 is M, What will be the value of 'n', if the equivalent mass of NH_4NO_3 is expressed as $\frac{M}{n}$?

58.



How many stereoisomer(s) is/are possible for the above compound.

59.



How many monochloro product(s) is/are formed in the above reaction?
(Consider minor as well as major products)

60. A solution contains 72 g of water and six moles of alcohol. What is the mole fraction of water in the solution?

Mathematics

PART – C

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. If $y = \tan \theta + \cot \theta$, $x = \tan \theta - \cot \theta$, where θ is a parameter then $y^3 \frac{d^2y}{dx^2}$ is equal to
- (A) 2
(B) -2
(C) 4
(D) -4
62. If the line $x \cos \theta + y \sin \theta = 2$ is the equation of a transverse common tangent to the circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 6\sqrt{3}x - 6y + 20 = 0$, then the value of θ is
- (A) $\frac{5\pi}{6}$
(B) $\frac{2\pi}{3}$
(C) $\frac{\pi}{3}$
(D) $\frac{\pi}{6}$
63. Let f be a twice differentiable function such that $f''(x) = 1 - f(x)$ where $f'(0) = f(0) = 1$.
Area bounded by the curve $y = f(x)$ and the x – axis from $x = 0$ to $x = \pi$ is
- (A) $2\pi + 4$
(B) $\pi + 2$
(C) $\pi + 4$
(D) $2\pi + 2$
64. The line $4x - 7y + 10 = 0$ intersects the parabola, $y^2 = 4x$ at the points A and B. The sum of the co – ordinate of the point of intersection of the tangents drawn at the point A and B is
- (A) 4
(B) 5
(C) 6
(D) 7

65. Consider the polynomial $P(x) = x^3 + ax^2 + bx + c$ where $a, b, c \in \{1, 2, \dots, 10\}$ then the probability that $P(x)$ is divisible by $x^2 + x + 1$ is
- (A) $\frac{1}{100}$
(B) $\frac{9}{1000}$
(C) $\frac{1}{1000}$
(D) $\frac{9}{100}$
66. Sum of the series $\frac{3}{4} + \frac{5}{36} + \frac{7}{144} + \dots$ upto n terms is equal to
- (A) $1 - \frac{1}{n^2}$
(B) $\frac{n(n+2)}{(n+1)^2}$
(C) $\frac{n^2 + 2n + 2}{(n+1)^2}$
(D) $1 + \frac{1}{n^2}$
67. If A is an idempotent matrix satisfying $(I - 0.4 A)^{-1} = I - \alpha A$, where I is unit matrix of the same order as that of A , then the value of α is:
- (A) $\frac{-1}{3}$
(B) $\frac{1}{3}$
(C) $\frac{-2}{3}$
(D) $\frac{2}{3}$
68. The unit digit of $17^{1983} + 11^{1983} - 7^{1983}$ is
- (A) 1
(B) 2
(C) 3
(D) 0

69. The equation $(\lambda - 1)x + (3\lambda + 1)y + 2\lambda z = 0$, $(\lambda - 1)x + (4\lambda - 2)y + (\lambda + 3)z = 0$ and $2x + (3\lambda + 1)y + 3(\lambda - 1)z = 0$ give non-trivial solution for some values of λ then the ratio $x : y : z$, when λ has smallest of these values is:
- (A) $3 : 2 : 1$
 (B) $1 : 3 : 2$
 (C) $1 : 3 : 1$
 (D) $1 : 1 : 1$
70. A game is played with fair cubic die which has one red side, two blue sides and three green sides. The result is the color of the top side after the dice is rolled. If the die is rolled repeatedly then the probability that second blue result occurs on or before the tenth roll can be expressed in the form of $\frac{3^p - 2^q}{3^r}$ where p, q, r are positive integers then the value of $p + r - q$ is
- (A) 9
 (B) 3
 (C) 7
 (D) 8
71. Two lines are drawn through $(3, 4)$ each of which makes angle of 45° with the line $x - y = 2$. Then area of the triangle formed by these lines is
- (A) 9
 (B) $\frac{9}{2}$
 (C) 2
 (D) $\frac{2}{9}$
72. The length of the major axis of the ellipse $(5x - 10)^2 + (5y + 15)^2 = \frac{(3x - 4y + 7)^2}{4}$ is
- (A) 10
 (B) $\frac{20}{3}$
 (C) $\frac{20}{7}$
 (D) 4
73. If $x^2 + px + 1$ is a factor of $ax^3 + bx + c$, then
- (A) $a^2 + c^2 = -ab$
 (B) $a^2 - c^2 = -ab$
 (C) $a^2 - c^2 = ab$
 (D) $a^2 + c^2 = ab$

74. If $\arg \frac{(z - (10 + 6i))}{(z - (4 + 2i))} = \frac{\pi}{4}$ then the perimeter of locus of z is:
- (A) $\sqrt{13} \left(\frac{\pi}{4} \right)$
(B) $3\sqrt{13} \pi$
(C) $3\sqrt{13} \left(\frac{\pi}{4} \right)$
(D) $3\sqrt{26} \left(\frac{\pi}{2} \right)$
75. Five digit numbers divisible by 9 are to be formed by using the digits 0, 1, 2, 3, 4, 5, 7 without repetition. The total numbers of such numbers is λ^3 , then λ is
- (A) 5
(B) 6
(C) 8
(D) 9
76. If $\sin \beta$ is the geometric mean between $\sin \alpha$ and $\cos \alpha$, then $\cos 2\beta$ is equal to
- (A) $2\sin^2 \left(\frac{\pi}{4} - \alpha \right)$
(B) $2\cos^2 \left(\frac{\pi}{4} - \alpha \right)$
(C) $2\sin^2 \left(\frac{\pi}{4} + \alpha \right)$
(D) None of these
77. If $2\sin^2 \left(\left(\frac{\pi}{2} \right) \cos^2 x \right) = 1 - \cos(\pi \sin 2x)$, $x \neq (2n+1)\frac{\pi}{2}$, $n \in \mathbb{I}$, then $\cos 2x$ is equal to
- (A) $\frac{1}{5}$
(B) $\frac{3}{5}$
(C) $\frac{4}{5}$
(D) 1

78. The value of $\cos^{-1}\left[\cot\left\{\sin^{-1}\left(\sqrt{\frac{2-\sqrt{3}}{4}}\right)+\cos^{-1}\left(\frac{\sqrt{12}}{4}\right)+\sec^{-1}(\sqrt{2})\right\}\right]$ is equal to
- (A) $\frac{\pi}{2}$
 (B) $\frac{\pi}{3}$
 (C) $\frac{\pi}{4}$
 (D) $\frac{\pi}{6}$
79. Given $f(g(x)) = 1 + x - 2x^2 + 5x^3$ and $f(x) = 1 + 3x \forall x \in \mathbb{R}$, then the value of $3g(1)$ is equal to
- (A) 5
 (B) 6
 (C) 4
 (D) 8
80. $\left[\lim_{n \rightarrow \infty} \sum_{r=0}^n \left({}^n C_r \cdot \frac{1}{(r+4) \cdot n^r}\right)\right]$ is equal to
 (where $[.]$ denotes greatest integer function):
- (A) 0
 (B) 3
 (C) 5
 (D) 7

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. If

$$\lim_{n \rightarrow \infty} n^k \int_0^{\frac{1}{n}} x^{x+k-1} dx$$

exists and is equal to $f(k)$, $k \in \mathbb{N}$, then the value of $\left[\frac{1}{f(5)}\right]$ is equal to (where $[.]$ denotes the greatest integer function)

82. The radius of circle which passes through the focus of the parabola $x^2 = 4y$ and touches it at point $(6, 9)$ is $k\sqrt{10}$, then k is equal to?

83. For the curve $y \ln x = x \ln y$, the intercept made by the normal at the point (e^2, e^2) on x-axis and y-axis are a and b respectively, then $\left| \frac{4a+b}{b} \right|$ is equal to
84. A normal to the hyperbola $\frac{x^2}{4} - \frac{y^2}{1} = 1$ has equal intercepts on positive x and y-axes. If this normal touches the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, then find $\frac{9}{25}(a^2 + b^2)$.
85. If $f(x) = \begin{cases} \tan^{-1}\left(\frac{ax+b}{c}\right) & 0 < |x| < \frac{1}{2} \\ 0 & x = \frac{1}{2} \\ \log(2+bx^2) & \frac{1}{2} < x < \frac{1}{\sqrt{2}} \end{cases}$ is differentiable in $x \in \left(0, \frac{1}{\sqrt{2}}\right)$, then $\left| \frac{a+b}{c} \right|$ is equal to
86. Suppose that all the terms of an arithmetic progression (A.P.) are natural numbers. If the ratio of the sum of the first seven terms to the sum of the first eleven terms is 6 : 11 and the seventh term lies in between 130 and 140, then the common difference of this A.P. is
87. The sum of the following infinite series $1 + \left(\frac{\sqrt{2}-1}{2\sqrt{2}}\right) + \left(\frac{3-2\sqrt{2}}{8}\right) + \left(\frac{5\sqrt{2}-7}{16\sqrt{2}}\right) + \frac{17-12\sqrt{2}}{64} + \dots \infty$ is
88. If $\int \frac{2x^7 + 3x^2}{x^{10} - 2x^5 + 1} dx = \frac{ax^3 + bx^8}{x^{10} - 2x^5 + 1} + c$ (where c is constant of integration), then $(a + b + 9)$ is equal to:
89. Rolle's Theorem holds for the function $f(x) = x^3 + bx^2 + cx$, $1 \leq x \leq 2$ at $x = 3$, then the value of $b + c$ is
90. Let $y = f(x)$ be a one – one function satisfying the differential equation $\frac{d^2y}{dx^2} + \frac{d^2x}{dy^2} = 0$ and $f'(0) = 1$, $f(1) = 2$, then $f(0)$ is: